

Executive Summary

Designing, Commercializing and Internationalizing Chinese Spherical Tokamaks — TUNEM Master’s Research Proposal

This proposal advances four core claims; the first two are technical and form the analytical heart of the thesis.

1. Reconstructing the design interface. Freidberg, Mangiarotti and Minervini’s *Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?* (Phys. Plasmas, 2015) shows that a reactor’s main parameters are fixed almost entirely by nuclear and engineering constraints, leaving the plasma to satisfy a small set of coupled limits. The thesis derives that “design interface” from first principles — the knobs (field B , shape and β , heating, pulse) and the operational limits (Troyon, Greenwald, kink, bootstrap) that bind them — rebuilding transparently what the specialist sources compress or assume.

2. Deriving how the spherical tokamak closes Freidberg’s constraints (the semi-original step). Freidberg derives that a conventional tokamak does not close: the required plasma current exceeds the kink limit, and the achievable bootstrap fraction (≈ 0.44) falls short of the value steady state requires (≈ 0.84). His paper offers high-field HTS magnets, the stellarator, and the fusion-fission hybrid as escapes — and does *not* develop the spherical tokamak. The thesis derives that the spherical tokamak closes *both* of his failure modes at once: low aspect ratio raises β (the route he treats sceptically) *and* structurally raises the bootstrap fraction (curing the very shortfall he identifies). This transparent derivation of the ST as one mathematically coherent route to close the problem his paper poses but leaves open is the proposal’s distinctive contribution.

3. An honest engineering-breakeven ledger. The thesis separates scientific breakeven from engineering breakeven and from economic viability, and places the spherical tokamak — and SUNIST-2 specifically — against the material walls that remain (high-field confinement, tritium breeding, neutron-tolerant first walls, continuous operation), claiming none of them solved.

4. From design interface to an integrated blueprint. Carried through to commercialization and internationalization, the same interface yields one accessible synthesis — Capabilities $\rightarrow$ Commercialization $\rightarrow$ International Markets — for the engineer-entrepreneur. This integration, absent from a literature that is either specialist or business-only, is what makes TUNEM’s engineering-and-management mission concrete, and it is the foundation of the author’s intended post-doctoral ventures. The entrepreneurial component is a minor share, treated at literature-review depth on public sources alone.

Primary sources. The analytical core rests on Freidberg, Mangiarotti & Minervini (2015); Freidberg, *Plasma Physics and Fusion Energy*; Wesson, *Tokamaks*; and Prof. Tan’s SUNIST-2 programme, whose stated aims — confinement at higher *magnetic field*, magnetic-reconnection heating, plasma-shaping and control, and repetitive pulsed operation — map directly onto the design knobs, the field knob above all.